Porto-Systemic Anastomosis

It comprises two components- Portal and Systemic veins. At connecting point referred to Porto-systemic anastomosis.

Portal vein: When vein begins with capillary and ends with capillary.

Superior mesenteric vein begins with capillary in gut wall and portal vein the continuation ends in another set of capillary in the liver.

NB: 8cm length, no valve, form like vein and ends like artery.

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Formation: Begins at the point of union of superior mesenteric vein and splenic vein behind the neck of pancreas at level of L2 vert. Continues upward with common bile duct to its right & gastro-duodenal artery on its left. Further up lies on free margin of lesser omentum where its posterior to two structures, common bile duct (right) and common hepatic artery (left).

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Close to porta hepatis divides into right & left branches.

Right br. Shorter, broader receives cystic vein before entering right lobe.

Left br. Longer, narrower, receives lig. teres, paraumbilical vein & connects with lig. venosum in the left lobe.

Portosystemic communications:

Lower end of oesophagus; oesophageal tributaries of left gastric veins (portal) anastomosis with oesophageal tributaries of accessory hemi-azygos vein (systemic). Blockage will cause varicose at lower end of oesophagus.

Anal canal; The superior rectal vein (portal) anastomosis with middle and inferior rectal veins (systemic).

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Umbilical area; Left branch of portal vein anastomosis with veins of anterior abdominal wall through paraumbilical veins. In portal obstruction veins around the umbilicus enlarges presenting *caput medusa*.

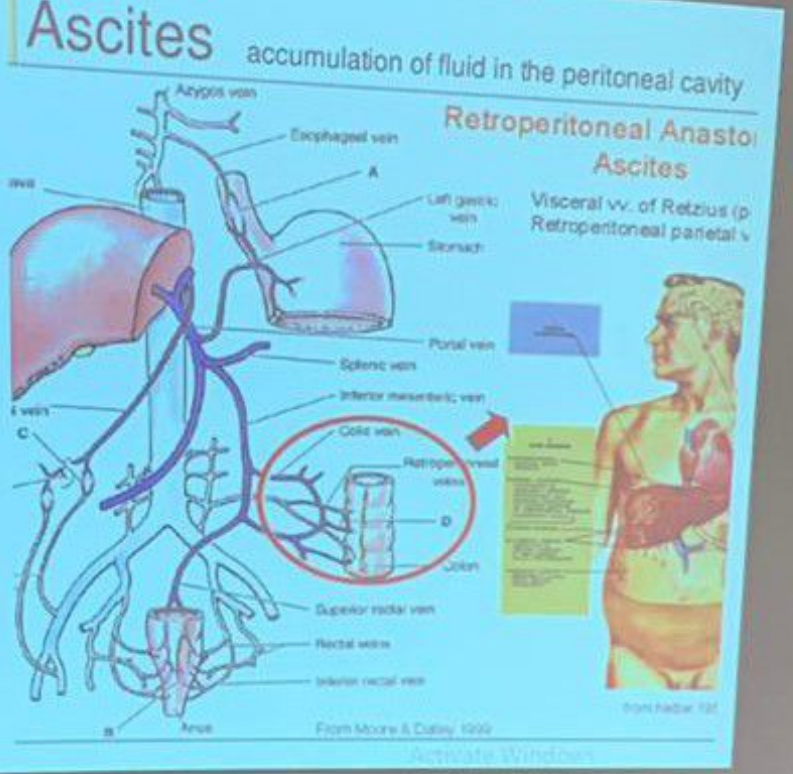
Liver; In a condition where the ductus venosus remains patent it connects the left branch of portal vein directly to inferior vena cava.



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Bare Area of liver; Hepatic venules of liver (portal) anastomoses with intercostal and phrenic veins (systemic).

Posterior abdominal wall; Veins of retroperitoneal organs like duodenum, ascending and descending colons (portal) anastomoses with retro-peritoneal veins of posterior abdominal wall and splenic capsule (systemic).



Functions of porto-systemic anastomoses:

- Provide alternative routes of venous blood circulation when there is a blockage in the liver or portal vein.
- Ensures that venous blood from the gastrointestinal tract still reaches the heart through the inferior vena cava without going through the liver.

Causes of portal hypertension; Can occur in the following ways;

1. Prehepatic; Congenital malformation or thrombosis or obliteration of portal vein.
2. Hepatic; Cirrhosis of liver in which there is significant obliteration of vascular bed of liver.
3. Posthepatic; Thrombosis of hepatic veins & formation of web caused by malignant cells.

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Effect of portal hypertension:

1. Congestive splenomegaly.
2. Leads to collateral circulation through the porto-systemic circulations.

What portal hypertension forms:

i. Caput medusa around umbilicus.

ii. Oesophageal varices around the lower end of oesophagus which may rupture causing haematemesis.

iii. Haemorrhoids in the anal canal which may lead to bleeding.

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